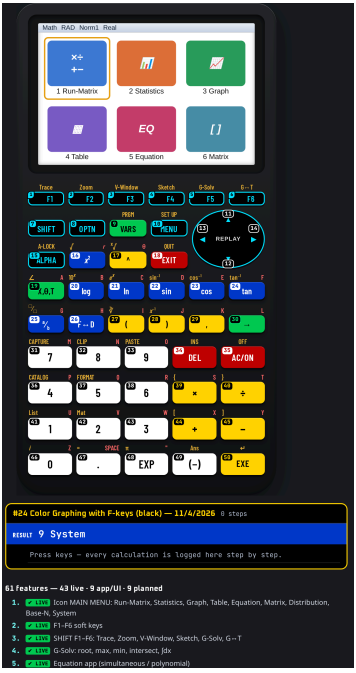
# U-45 CALCULATORS

## User Manual

### Calculator #24 Color Graphing with F-keys (black)

Graphing - colour




Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

# About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#24 — Color Graphing with F-keys (black)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

## How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 F1]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

## Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

> This model starts on its **main menu**. Press [#41 1] to open the calculation app first; the examples include this step.

## Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills

10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)
12. Appendix C — Glossary
13. Appendix D — Model notes & limits
14. Appendix E — Quick-reference card

# 1 · Getting started

Open U-45 CALCULATORS, choose **#24** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

### Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

### High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

### Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

### Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

### Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

### Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

### Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

### Tooltips

Hover over (or long-press) a key to see "#number name".

### Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

### History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

### Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

### Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

### Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

### Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

### Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

## Colour groups on this calculator

| Colour | Group | Keys on this model |
|---|---|---|
| num | Digits & number entry | #31, #32, #33, #36, #37, #38, #41, #42, #43, #46, #47, #48, #49 |
| op | Operators & execute | #17, #27, #28, #29, #39, #40, #44, #45, #50 |
| fn | Functions | #16, #20, #21, #22, #23, #24, #25, #26 |
| mem | Memory, variables & registers | #9, #19, #30 |
| clr | Clear, delete & power | #18, #34, #35 |
| mod | Shift, modes, menus & navigation | #1, #2, #3, #4, #5, #6, #7, #8, #10, #11, #12, #13, #14, #15 |

# 2 · Features

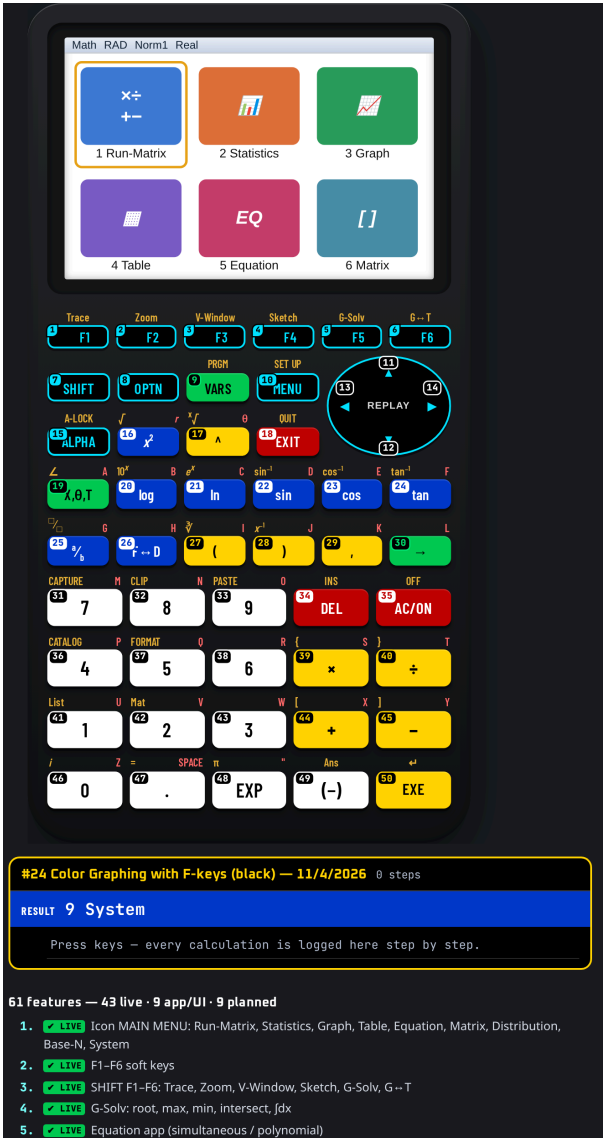
**61 features: 43 live on the keys, 9 app/UI features, 9 planned (documented but not yet working — they are clearly marked and never used in examples).**

1. LIVE **Icon MAIN MENU: Run-Matrix, Statistics, Graph, Table, Equation, Matrix, Distribution, Base-N, System**
2. LIVE **F1–F6 soft keys**
3. LIVE **SHIFT F1–F6: Trace, Zoom, V-Window, Sketch, G-Solv, G ↔ T**
4. LIVE **G-Solv: root, max, min, intersect, ∫dx**
5. LIVE **Equation app (simultaneous / polynomial)**
6. LIVE **Statistics lists with regressions**
7. LIVE **Distribution app**
8. LIVE **Base-N app**
9. LIVE **OPTN / VARS menus**
10. LIVE **Variables A–Z, r, θ**
11. LIVE **Fraction template ▭/▭ with exact fraction results**
12. LIVE **Mixed-number entry (a b/c) and improper ↔ mixed conversion**
13. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
14. LIVE **Square root √**
15. LIVE **Cube root $\sqrt[3]{}$**
16. LIVE **Any-index root $\sqrt[x]{y}$**
17. LIVE **Square $x^2$**
18. LIVE **Reciprocal $x^{-1}$ (1/x)**
19. LIVE **Power $x^y$ / ^ with right-to-left chaining (2^3^2 = 512)**

20. LIVE **Common logarithm $\log_{10}$**
21. LIVE **Natural logarithm ln**
22. LIVE **Antilog $10^x$**
23. LIVE **Exponential $e^x$**
24. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
25. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
26. LIVE **π constant to full internal precision**
27. LIVE **Scientific-notation entry $\times 10^x$ / EXP / EEX**
28. LIVE **Sign change (−) / +/− / CHS**
29. LIVE **Ans recall of the last result; operators auto-insert Ans**
30. LIVE **Store a value into memory / a variable (STO, sto►, →, Min)**
31. LIVE **Named variables (A–F, X, Y, M … or A–Z) usable inside expressions**
32. LIVE **TABLE — tabulate f(x) over start / end / step**
33. LIVE **Statistics: data entry, n, $\bar{x}$, $\sigma$, s, $\Sigma x$, $\Sigma x^2$**
34. LIVE **Two-variable statistics and linear regression (a, b, r)**
35. LIVE **Matrix mode: define, add, multiply, det, transpose, identity**
36. LIVE **Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular**
37. LIVE **BASE-N: DEC / HEX / BIN / OCT conversions**
38. LIVE **Bitwise logic AND / OR / XOR / XNOR / NOT**
39. LIVE **Backspace / DEL editing of the current entry**
40. LIVE **INS insert mode for editing in the middle of an expression**
41. LIVE **Replay ▲ ▼ — scroll back to previous calculations and re-edit them**
42. LIVE **CATALOG — alphabetical list of every function**

43. LIVE Probability distributions (normal, binomial, Poisson) and PRB menu
44. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
45. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
46. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
47. APP/UI Show / hide key numbers with one button (remembered between sessions)
48. APP/UI Copy the current display text to the clipboard
49. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
50. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
51. APP/UI Reference photo viewer and tool-tip name on every key
52. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
53. PLANNED Programs
54. PLANNED e-Activity
55. PLANNED Spreadsheet
56. PLANNED CAPTURE / CLIP / PASTE
57. PLANNED Graph image export (PNG)
58. PLANNED Shareable function list link
59. PLANNED Dark screen theme
60. PLANNED Speech read-out of values
61. PLANNED Exam mode

# 3 · Numbered key map




Calculator #24 with key numbers shown (Hi-Contrast colours).

## Key table

**"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.**

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 1 | F1 | Trace | · | · | mod |
| 2 | F2 | Zoom | · | · | mod |
| 3 | F3 | V-Window | · | · | mod |
| 4 | F4 | Sketch | · | · | mod |
| 5 | F5 | G-Solv | · | · | mod |
| 6 | F6 | G ↔ T | · | · | mod |
| 7 | SHIFT | · | · | · | mod |
| 8 | OPTN | · | · | · | mod |
| 9 | VARS | PRGM | · | · | mem |
| 10 | MENU | SET UP | · | · | mod |
| 11 | ▲ | · | · | · | mod |
| 12 | ▼ | · | · | · | mod |
| 13 | ◀ | · | · | · | mod |
| 14 | ▶ | · | · | · | mod |
| 15 | ALPHA | A-LOCK | · | · | mod |
| 16 | $x^2$ | √ | r | · | fn |
| 17 | ^ | x√ | θ | · | op |
| 18 | EXIT | QUIT | · | · | clr |
| 19 | X,θ,T | ∠ | A | · | mem |
| 20 | log | 10x | B | · | fn |
| 21 | ln | ex | C | · | fn |
| 22 | sin | $\sin^{-1}$ | D | · | fn |
| 23 | cos | $\cos^{-1}$ | E | · | fn |

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 24 | tan | $\tan^{-1}$ | F | · | fn |
| 25 | a/b | □/□ | G | · | fn |
| 26 | F ↔ D | · | H | · | fn |
| 27 | ( | $\sqrt[3]{}$ | I | · | op |
| 28 | ) | $x^{-1}$ | J | · | op |
| 29 | , | · | K | · | op |
| 30 | → | · | L | · | mem |
| 31 | 7 | CAPTURE | M | · | num |
| 32 | 8 | CLIP | N | · | num |
| 33 | 9 | PASTE | O | · | num |
| 34 | DEL | INS | · | · | clr |
| 35 | AC/ON | OFF | · | · | clr |
| 36 | 4 | CATALOG | P | · | num |
| 37 | 5 | FORMAT | Q | · | num |
| 38 | 6 | · | R | · | num |
| 39 | × | { | S | · | op |
| 40 | ÷ | } | T | · | op |
| 41 | 1 | List | U | · | num |
| 42 | 2 | Mat | V | · | num |
| 43 | 3 | · | W | · | num |
| 44 | + | [ | X | · | op |
| 45 | − | ] | Y | · | op |
| 46 | 0 | i | Z | · | num |
| 47 | . | = | SPACE | · | num |
| 48 | EXP | π | " | · | num |
| 49 | (−) | Ans | · | · | num |
| 50 | EXE | ↵ | · | · | op |

# 4 · Key-by-key reference

**Every key on calculator #24, in number order, with every function it carries in this app.**

### #1 F1 `Shift, modes, menus & navigation`

**Primary:** F1 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** Trace — TRACE — move along a graph

### #2 F2 `Shift, modes, menus & navigation`

**Primary:** F2 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** Zoom — ZOOM — zoom presets

### #3 F3 `Shift, modes, menus & navigation`

**Primary:** F3 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** V-Window — V-Window — graph view settings

### #4 F4 `Shift, modes, menus & navigation`

**Primary:** F4 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** Sketch — DRAW menu

### #5 F5 `Shift, modes, menus & navigation`

**Primary:** F5 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** G-Solv — CALC — graph analysis (zero, min, max, intersect)

### #6 F6 `Shift, modes, menus & navigation`

**Primary:** F6 soft key

**shifted (2nd / f / SHIFT layer) (press #7 first):** G ↔ T — TABLE — table of values

### #7 SHIFT `Shift, modes, menus & navigation`

**Primary:** SHIFT / 2nd — the next key uses its upper (shift-coloured) function

### #8 OPTN `Shift, modes, menus & navigation`

**Primary:** OPTN — options for the current mode

### #9 VARS `Memory, variables & registers`

**Primary:** VARS — variables menu

**shifted (2nd / f / SHIFT layer) (press #7 first):** PRGM — PRGM — programs

### #10 MENU `Shift, modes, menus & navigation`

**Primary:** MENU — main menu

**shifted (2nd / f / SHIFT layer) (press #7 first):** SET UP — Graph / app mode

## #11 ▲ `Shift, modes, menus & navigation`

**Primary:** Cursor ▲ — up / previous history line / previous item

## #12 ▼ `Shift, modes, menus & navigation`

**Primary:** Cursor ▼ — down / next history line / next item

## #13 ◀ `Shift, modes, menus & navigation`

**Primary:** Cursor ◀ — move left in the line

## #14 ▶ `Shift, modes, menus & navigation`

**Primary:** Cursor ▶ — move right in the line

## #15 ALPHA `Shift, modes, menus & navigation`

**Primary:** ALPHA — the next key types its letter/variable (red/green label)

**shifted (2nd / f / SHIFT layer) (press #7 first):** A-LOCK — ALPHA — the next key types its letter/variable (red/green label)

## #16 $x^2$ `Functions`

**Primary:** Square $x^2$

**shifted (2nd / f / SHIFT layer) (press #7 first):** $\sqrt{}$ — Square root $\sqrt{}$

**alpha layer (press #15 first):** r — Variable r (store with STO/ →, recall by name)

**Keyboard:** q

## #17 ^ Operators & execute

**Primary:** Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

**shifted (2nd / f / SHIFT layer) (press #7 first):** $x\sqrt{}$ — n-th root $\sqrt[x]{y}$

**alpha layer (press #15 first):** θ — Variable θ (store with STO/ →, recall by name)

**Keyboard:** ^

## #18 EXIT Clear, delete & power

**Primary:** QUIT — leave the current screen / app

**shifted (2nd / f / SHIFT layer) (press #7 first):** QUIT — QUIT — leave the current screen / app

## #19 X,θ,T Memory, variables & registers

**Primary:** The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving

**alpha layer (press #15 first):** A — Variable A (store with STO/ →, recall by name)

## #20 log Functions

**Primary:** Common logarithm $\log_{10}$

**shifted (2nd / f / SHIFT layer) (press #7 first):** 10x — $10^x$ — antilog

**alpha layer (press #15 first):** B — Variable B (store with STO/ →, recall by name)

**Keyboard:** l

## #21 ln Functions

**Primary:** Natural logarithm ln

**shifted (2nd / f / SHIFT layer) (press #7 first):** ex — $e^x$ — exponential

**alpha layer (press #15 first):** C — Variable C (store with STO/ →, recall by name)

**Keyboard:** n

## #22 sin Functions

**Primary:** Sine (uses the current angle unit)

**shifted (2nd / f / SHIFT layer) (press #7 first):** $\sin^{-1}$ — Inverse sine $\sin^{-1}$

**alpha layer (press #15 first):** D — Variable D (store with STO/ →, recall by name)

**Keyboard:** s

## #23 cos Functions

**Primary:** Cosine (current angle unit)

**shifted (2nd / f / SHIFT layer) (press #7 first):** $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

**alpha layer (press #15 first):** E — Variable E (store with STO/ →, recall by name)

**Keyboard:** c

## #24 tan Functions

**Primary:** Tangent (current angle unit)

**shifted (2nd / f / SHIFT layer) (press #7 first):** $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

**alpha layer (press #15 first):** F — Variable F (store with STO/ →, recall by name)

**Keyboard:** t

## #25 a/b Functions

**Primary:** Fraction template ▭/▭ (numerator over denominator)

**shifted (2nd / f / SHIFT layer) (press #7 first):** □/□ — Mixed-number template (whole, numerator, denominator)

**alpha layer (press #15 first):** G — Variable G (store with STO/ →, recall by name)

## #26 F ↔ D Functions

**Primary:** S⇔D — toggle the result between exact (fraction, √, π) and decimal

**alpha layer (press #15 first):** H — Variable H (store with STO/ →, recall by name)

## #27 ( Operators & execute

**Primary:** Opening bracket (

**shifted (2nd / f / SHIFT layer) (press #7 first):** $\sqrt[3]{}$ — Cube root $\sqrt[3]{}$

**alpha layer (press #15 first):** I — Variable I (store with STO/ →, recall by name)

**Keyboard:** [(]

## #28 ) Operators & execute

**Primary:** Closing bracket ). Closing brackets at the end of a line may be omitted

**shifted (2nd / f / SHIFT layer) (press #7 first):** $x^{-1}$ — Reciprocal $x^{-1}$ (1/x)

**alpha layer (press #15 first):** J — Variable J (store with STO/ →, recall by name)

**Keyboard:** [)]

**#29** , Operators & execute

**Primary:** Comma / argument separator (between function arguments, list items or coordinates)

**alpha layer (press #15 first):** K — Variable K (store with STO/ →, recall by name)

**Keyboard:** ,

**#30** → Memory, variables & registers

**Primary:** STO▶ ( → ) — store into a variable

**alpha layer (press #15 first):** L — Variable L (store with STO/ →, recall by name)

**#31** 7 Digits & number entry

**Primary:** Digit 7

**alpha layer (press #15 first):** M — Variable M (store with STO/ →, recall by name)

**Keyboard:** 7

**#32** 8 Digits & number entry

**Primary:** Digit 8

**alpha layer (press #15 first):** N — Variable N (store with STO/ →, recall by name)

**Keyboard:** 8

### #33 9 `Digits & number entry`

**Primary:** Digit 9

**alpha layer (press #15 first):** 0 — Variable 0 (store with STO/ →, recall by name)

**Keyboard:** 9

### #34 DEL `Clear, delete & power`

**Primary:** DEL — delete the character at the cursor

**shifted (2nd / f / SHIFT layer) (press #7 first):** INS — INS — insert mode at the cursor

**Keyboard:** Backspace Delete

### #35 AC/ON `Clear, delete & power`

**Primary:** ON / AC — power on, or clear all

**shifted (2nd / f / SHIFT layer) (press #7 first):** OFF — OFF — power off (state is lost; press ON to restart)

### #36 4 `Digits & number entry`

**Primary:** Digit 4

**shifted (2nd / f / SHIFT layer) (press #7 first):** CATALOG — CATALOG — list of all commands

**alpha layer (press #15 first):** P — Variable P (store with STO/ →, recall by name)

**Keyboard:** 4

## #37 5 Digits & number entry

**Primary:** Digit 5

**shifted (2nd / f / SHIFT layer) (press #7 first):** FORMAT — FORMAT — choose number format (Fix/Sci/Norm)

**alpha layer (press #15 first):** Q — Variable Q (store with STO/ →, recall by name)

**Keyboard:** 5

## #38 6 Digits & number entry

**Primary:** Digit 6

**alpha layer (press #15 first):** R — Variable R (store with STO/ →, recall by name)

**Keyboard:** 6

## #39 × Operators & execute

**Primary:** Multiplication ×

**shifted (2nd / f / SHIFT layer) (press #7 first):** { — Types "{"

**alpha layer (press #15 first):** S — Variable S (store with STO/ →, recall by name)

**Keyboard:** * X

## #40 ÷ Operators & execute

**Primary:** Division ÷

**shifted (2nd / f / SHIFT layer) (press #7 first):** } — Types "}"

**alpha layer (press #15 first):** T — Variable T (store with STO/ →, recall by name)

**Keyboard:** /

**#41** **1** `Digits & number entry`

**Primary:** Digit 1

**shifted (2nd / f / SHIFT layer) (press #7 first):** List — LIST menu

**alpha layer (press #15 first):** U — Variable U (store with STO/ →, recall by name)

**Keyboard:** 1

**#42** **2** `Digits & number entry`

**Primary:** Digit 2

**shifted (2nd / f / SHIFT layer) (press #7 first):** Mat — MATRIX menu

**alpha layer (press #15 first):** V — Variable V (store with STO/ →, recall by name)

**Keyboard:** 2

**#43** **3** `Digits & number entry`

**Primary:** Digit 3

**alpha layer (press #15 first):** W — Variable W (store with STO/ →, recall by name)

**Keyboard:** 3

**#44** **+** `Operators & execute`

**Primary:** Addition +

**shifted (2nd / f / SHIFT layer) (press #7 first):** [ — Types "["

**alpha layer (press #15 first):** X — Variable X (store with STO/ →, recall by name)

**Keyboard:** +

**#45** **–** `Operators & execute`

**Primary:** Subtraction –

**shifted (2nd / f / SHIFT layer) (press #7 first):** ] — Types "]"

**alpha layer (press #15 first):** Y — Variable Y (store with STO/ →, recall by name)

**Keyboard:** -

**#46** **0** `Digits & number entry`

**Primary:** Digit 0

**shifted (2nd / f / SHIFT layer) (press #7 first):** i — Imaginary unit i (complex numbers)

**alpha layer (press #15 first):** Z — Variable Z (store with STO/ →, recall by name)

**Keyboard:** 0

**#47** **.** `Digits & number entry`

**Primary:** Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

**shifted (2nd / f / SHIFT layer) (press #7 first):** = — Equals — complete the calculation

**Keyboard:** .

## #48 EXP Digits & number entry

**Primary:** ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

**shifted (2nd / f / SHIFT layer) (press #7 first):** π — π (3.14159265…) as an exact constant

**Keyboard:** e

## #49 (−) Digits & number entry

**Primary:** Negative sign (−) — makes the next number negative. Different from the subtraction key

**shifted (2nd / f / SHIFT layer) (press #7 first):** Ans — Ans — recall the last answer into the current expression

## #50 EXE Operators & execute

**Primary:** Execute / = — evaluate the line and show the result (EXE / ENTER / =)

**shifted (2nd / f / SHIFT layer) (press #7 first):** ↵ — Execute / = — evaluate the line and show the result (EXE / ENTER / =)

**Keyboard:** Enter =

# 5 · Worked examples

**18 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.**

### Example 5.1 — Order of operations

**Calculate 12 + 3 × 4. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #41 1 → #42 2 → #44 + → #43 3 → #39 × → #36 4 → #50 EXE

**Display shows** 24 **— expected 24. ✔ verified on the app.**

*Multiplication is done before addition, so the calculator holds the + until the × part is finished.*

### Example 5.2 — Brackets first

**Calculate (12 + 8) ÷ 4. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #27 ( → #41 1 → #42 2 → #44 + → #32 8 → #28 ) → #40 ÷ → #36 4 → #50 EXE

**Display shows** 5 **— expected 5. ✔ verified on the app.**

*Brackets force the sum to be worked out before the division.*

### Example 5.3 — Two brackets multiplied

**Calculate (3 + 4) × (9 − 2). Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #27 ( → #43 3 → #44 + → #36 4 → #28 ) → #39 × → #27 ( → #33 9 → #45 − → #42 2 → #28 ) → #50 EXE

**Display shows** 49 **— expected 49. ✔ verified on the app.**

*Both brackets are evaluated first, then multiplied.*

### Example 5.4 — Squares

**Calculate $12^2$. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #41 1 → #42 2 → #16 $x^2$ → #50 EXE

**Display shows** 144 **— expected 144. ✔ verified on the app.**

*$x^2$ squares the value — quicker and safer than typing the number twice.*

### Example 5.5 — Square roots

**Calculate √144. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #7 SHIFT → #16 √ → #41 1 → #36 4 → #36 4 → #50 EXE

**Display shows** 12 **— expected 12. ✔ verified on the app.**

*The square-root key returns the principal (positive) root.*

### Example 5.6 — Powers

**Calculate 2^10. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #42 2 → #17 ^ → #41 1 → #46 0 → #50 EXE

**Display shows** 1024 **— expected 1024. ✔ verified on the app.**

*The power key raises the first number to the second.*

### Example 5.7 — Reciprocals

**Calculate 1 ÷ 0.25 ($x^{-1}$). Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #46 0 → #47 . → #42 2 → #37 5 → #7 SHIFT → #28 $x^{-1}$ → #50 EXE

**Display shows** 4 **— expected 4. ✔ verified on the app.**

*$x^{-1}$ (1/x) gives the reciprocal — handy for resistors in parallel or rates.*

### Example 5.8 — Trigonometry (degrees)

**Calculate cos 0° (Deg mode). Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #23 cos → #46 0 → #28 ) → #50 EXE

**Display shows** 1 **— expected 1. ✔ verified on the app.**

*Trig functions use the current angle unit; these answers assume degrees (the start-up setting).*

### Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#41 1 *(open the calculation app from the main menu)* → #20 log → #41 1 → #46 0 → #46 0 → #46 0 → #28 ) → #50 EXE

Display shows 3 — expected 3. ✔ verified on the app.

*log is base-10: it counts the zeros of a power of ten.*

### Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#41 1 *(open the calculation app from the main menu)* → #21 ln → #41 1 → #28 ) → #50 EXE

Display shows 0 — expected 0. ✔ verified on the app.

*ln is the natural (base-e) logarithm.*

### Example 5.11 — Negative numbers

Calculate **−5 + 12**. Press **Reset** first.

#41 1 *(open the calculation app from the main menu)* → #49 (−) → #37 5 → #44 + → #41 1 → #42 2 → #50 EXE

Display shows 7 — expected 7. ✔ verified on the app.

*Use the sign key for a negative number — not the subtraction key.*

### Example 5.12 — Scientific notation

**Calculate 6×10^2 × 3. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #38 6 → #48 EXP → #42 2 → #39 × → #43 3 → #50 EXE

**Display shows** `1800` **— expected 1800. ✔ verified on the app.**

*The exponent key enters ×10ⁿ without typing × 10 ^.*

### Example 5.13 — Area of a circle πr²

**Calculate π × 2.5². Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #7 SHIFT → #48 π → #39 × → #42 2 → #47 . → #37 5 → #16 $x^2$ → #50 EXE

**Display shows** `19.63495408` **— expected 19.63495408. ✔ verified on the app.**

*π is stored to full precision — never type 3.14.*

### Example 5.14 — Continue from the answer

Calculate **100 × 4 = , then Ans ÷ 2.** Press **Reset** first.

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#46` 0 → `#46` 0 → `#39` × → `#36` 4 → `#50` EXE → `#40` ÷ → `#42` 2 → `#50` EXE

Display shows `200` — expected 200. ✔ verified on the app.

*Pressing an operator straight after a result continues from that answer (Ans).*

### Example 5.15 — Average of three numbers

Calculate **(4 + 8 + 15) ÷ 3.** Press **Reset** first.

`#41` 1 *(open the calculation app from the main menu)* → `#27` ( → `#36` 4 → `#44` + → `#32` 8 → `#44` + → `#41` 1 → `#37` 5 → `#28` ) → `#40` ÷ → `#43` 3 → `#50` EXE

Display shows `9` — expected 9. ✔ verified on the app.

*Add up the values, then divide by how many there are.*

### Example 5.16 — Mixed operations

**Calculate 50 − 4 × 6 + 7. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #37 5 → #46 0 → #45 − → #36 4 → #39 × → #38 6 → #44 + → #31 7 → #50 EXE

**Display shows** 33 **— expected 33. ✔ verified on the app.**

*Only the × part is done first; the − and + are then worked left to right.*

### Example 5.17 — Dividing twice

**Calculate 360 ÷ 4 ÷ 9. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #43 3 → #38 6 → #46 0 → #40 ÷ → #36 4 → #40 ÷ → #33 9 → #50 EXE

**Display shows** 10 **— expected 10. ✔ verified on the app.**

*Division is worked left to right: the first quotient is divided again.*

### Example 5.18 — Brackets inside a longer formula

**Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.**

#41 1 *(open the calculation app from the main menu)* → #27 ( → #27 ( → #43 3 → #44 + → #37 5 → #28 ) → #39 × → #36 4 → #45 − → #42 2 → #28 ) → #40 ÷ → #41 1 → #46 0 → #50 EXE

**Display shows** 3 **— expected 3. ✔ verified on the app.**

*The innermost brackets are worked out first, then the outer brackets, then the division.*

# 6 · Tricks & hacks

### Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

### Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

### Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

### Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

### Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

### Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

### Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

### Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

### Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

### Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

### Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

### Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

### Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

### Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

### Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

### Hack 16: Hidden functions on this model

63 extra functions sit on shifted layers. Press the layer key once, then the key: for example #7 then #1 → Trace; #7 then #2 → Zoom; #7 then #3 → V-Window; #7 then #4 → Sketch.

### Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #50.

### Hack 18: Type faster on a computer

28 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

### Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

### Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

### Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

# 7 · Keyboard shortcuts

**With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.**

| Keyboard | # | Key | Does |
|---|---|---|---|
| q | #16 | $x^2$ | Square $x^2$ |
| ^ | #17 | ^ | Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512) |
| l | #20 | log | Common logarithm $\log_{10}$ |
| n | #21 | ln | Natural logarithm ln |
| s | #22 | sin | Sine (uses the current angle unit) |
| c | #23 | cos | Cosine (current angle unit) |
| t | #24 | tan | Tangent (current angle unit) |
| ( | #27 | ( | Opening bracket ( |
| ) | #28 | ) | Closing bracket ). Closing brackets at the end of a line may be omitted |
| , | #29 | , | Comma / argument separator (between function arguments, list items or coordinates) |
| 7 | #31 | 7 | Digit 7 |
| 8 | #32 | 8 | Digit 8 |
| 9 | #33 | 9 | Digit 9 |
| Backspace Delete | #34 | DEL | DEL — delete the character at the cursor |
| 4 | #36 | 4 | Digit 4 |
| 5 | #37 | 5 | Digit 5 |
| 6 | #38 | 6 | Digit 6 |
| * x | #39 | × | Multiplication × |
| / | #40 | ÷ | Division ÷ |
| 1 | #41 | 1 | Digit 1 |

| | | | |
|---|---|---|---|
| 2 | #42 | 2 | Digit 2 |
| 3 | #43 | 3 | Digit 3 |
| + | #44 | + | Addition + |
| - | #45 | − | Subtraction − |
| 0 | #46 | 0 | Digit 0 |
| . | #47 | . | Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored |
| e | #48 | EXP | ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23 |
| Enter = | #50 | EXE | Execute / = — evaluate the line and show the result (EXE / ENTER / =) |

## App shortcuts

| | |
|---|---|
| “# On / # Off” | show / hide key numbers |
| “Classic / Hi-Contrast” | switch key colours |
| “Copy” | copy display text |
| “Fit / 1:1” | scale to screen / design size |
| “Reset” | restart this calculator |
| Jump list | switch to another of the 45 calculators |

# 8 · Common mistakes

**Mistake 1: (–) vs –**

Use the negative key for negative numbers.

**Mistake 2: Radian mode**

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

**Mistake 3: Implicit multiplication**

2/3x may be read as 2/(3x) depending on the model; use brackets.

**Mistake 4: Using – instead of (–)**

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

**Mistake 5: Wrong angle unit**

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

**Mistake 6: $-3^2$ vs $(-3)^2$**

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

### Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

### Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

### Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

### Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

### Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

### Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

# 9 · Practice drills

## Set A — Calculation drills (92)

**Work each one on calculator #24, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.**

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. $25^2$ = _____

21. $7^2$ = _____

22. $1.5^2$ = _____

23. $19^2$ = _____

24. $33^2$ = _____

25. $0.5^2$ = _____

26. $125^2$ = _____

27. $64^2$ = _____

28. √625 = _____

29. √1024 = _____

30. √9801 = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

**37.** 2^0.5 = _____

**38.** 1 ÷ 0.2 ($x^{-1}$) = _____

**39.** log 100 = _____

**40.** log 10000 = _____

**41.** ln 10 = _____

**42.** ln 2 = _____

**43.** −20 + 7 = _____

**44.** −3 + 3 = _____

**45.** −100 + 250 = _____

**46.** −8 + 1 = _____

**47.** 2×10^3 × 4 = _____

**48.** 1.5×10^4 × 2 = _____

**49.** 3×10^5 × 3 = _____

**50.** 4.5×10^2 × 2 = _____

**51.** π × $1.2^2$ = _____

**52.** 15 × 6 = , then Ans ÷ 9 = _____

**53.** 7.5 × 8 = , then Ans ÷ 3 = _____

**54.** 250 × 12 = , then Ans ÷ 100 = _____

**55.** (12 + 15 + 18) ÷ 3 = _____

**56.** (70 + 85 + 91) ÷ 3 = _____

57. (2.5 + 3.5 + 6) ÷ 3 = _____

58. (100 + 200 + 600) ÷ 3 = _____

59. 100 − 3 × 12 + 9 = _____

60. 18 − 2 × 5 + 40 = _____

61. 7.5 − 1.5 × 3 + 2 = _____

62. 64 − 8 × 4 + 10 = _____

63. 300 − 25 × 4 + 55 = _____

64. 9 − 3 × 3 + 1 = _____

65. 45 − 5 × 5 + 15 = _____

66. 1000 ÷ 8 ÷ 5 = _____

67. 96 ÷ 2 ÷ 3 = _____

68. 7200 ÷ 60 ÷ 60 = _____

69. 500 ÷ 4 ÷ 25 = _____

70. 81 ÷ 3 ÷ 3 = _____

71. ((10 + 2) × 3 − 6) ÷ 6 = _____

72. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

73. ((20 + 30) × 2 − 25) ÷ 3 = _____

74. ((7 + 8) × 9 − 15) ÷ 12 = _____

75. 12 + 3 × 4 = _____

76. (12 + 8) ÷ 4 = _____

77. (3 + 4) × (9 − 2) = _____

78. $12^2$ = _____

79. √144 = _____

80. 2^10 = _____

81. 1 ÷ 0.25 ($x^{-1}$) = _____

82. cos 0° (Deg mode) = _____

83. log 1000 = _____

84. ln 1 = _____

85. −5 + 12 = _____

86. 6×10^2 × 3 = _____

87. π × $2.5^2$ = _____

88. 100 × 4 = , then Ans ÷ 2 = _____

89. (4 + 8 + 15) ÷ 3 = _____

90. 50 − 4 × 6 + 7 = _____

91. 360 ÷ 4 ÷ 9 = _____

92. ((3 + 5) × 4 − 2) ÷ 10 = _____

## Set B — Key-finding drills (103)

**Without looking at the table, find each key on the picture. Answers in Appendix A.**

1. Which key is “F1” (F1 soft key)? _____
2. Which key is “F2” (F2 soft key)? _____

3. Which key is “F3” (F3 soft key)? _____

4. Which key is “F4” (F4 soft key)? _____

5. Which key is “F5” (F5 soft key)? _____

6. Which key is “F6” (F6 soft key)? _____

7. Which key is “SHIFT” (SHIFT / 2nd)? _____

8. Which key is “OPTN” (OPTN)? _____

9. Which key is “VARS” (VARS)? _____

10. Which key is “MENU” (MENU)? _____

11. Which key is “▲” (Cursor ▲)? _____

12. Which key is “▼” (Cursor ▼)? _____

13. Which key is “◀” (Cursor ◀)? _____

14. Which key is “▶” (Cursor ▶)? _____

15. Which key is “ALPHA” (ALPHA)? _____

16. Which key is “$x^2$” (Square $x^2$)? _____

17. Which key is “^” (Power $x^y$ / ^)? _____

18. Which key is “EXIT” (QUIT)? _____

19. Which key is “X,θ,T” (The graphing variable)? _____

20. Which key is “log” (Common logarithm $\log_{10}$)? _____

21. Which key is “ln” (Natural logarithm ln)? _____

22. Which key is “sin” (Sine)? _____

23. Which key is “cos” (Cosine)? _____

24. Which key is “tan” (Tangent)? _____

25. Which key is “a⁄b” (Fraction template ▭/▭)? _____

26. Which key is “F ↔ D” (S⇔D)? _____

27. Which key is “(” (Opening bracket)? _____

28. Which key is “)” (Closing bracket ). Closing brackets at the end of a line may be omitted)? _____

29. Which key is “,” (Comma / argument separator)? _____

30. Which key is “ → ” (STO▶)? _____

31. Which key is “DEL” (DEL)? _____

32. Which key is “AC/ON” (ON / AC)? _____

33. Which key is “×” (Multiplication ×)? _____

34. Which key is “÷” (Division ÷)? _____

35. Which key is “+” (Addition +)? _____

36. Which key is “−” (Subtraction −)? _____

37. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

38. Which key is “EXP” (×10ˣ / EXP / EE)? _____

39. Which key is “(−)” (Negative sign)? _____

40. Which key is “EXE” (Execute / =)? _____

41. How do you reach “Trace”? _____

**42. How do you reach “Zoom”? _____**

**43. How do you reach “V-Window”? _____**

**44. How do you reach “Sketch”? _____**

**45. How do you reach “G-Solv”? _____**

**46. How do you reach “G ↔ T”? _____**

**47. How do you reach “PRGM”? _____**

**48. How do you reach “SET UP”? _____**

**49. How do you reach “A-LOCK”? _____**

**50. How do you reach “√”? _____**

**51. How do you reach “r”? _____**

**52. How do you reach “x√”? _____**

**53. How do you reach “θ”? _____**

**54. How do you reach “QUIT”? _____**

**55. How do you reach “A”? _____**

**56. How do you reach “10x”? _____**

**57. How do you reach “B”? _____**

**58. How do you reach “ex”? _____**

**59. How do you reach “C”? _____**

**60. How do you reach “$\sin^{-1}$”? _____**

**61. How do you reach “D”? _____**

**62. How do you reach "cos$^{-1}$"? _____**

**63. How do you reach "E"? _____**

**64. How do you reach "tan$^{-1}$"? _____**

**65. How do you reach "F"? _____**

**66. How do you reach "□/□"? _____**

**67. How do you reach "G"? _____**

**68. How do you reach "H"? _____**

**69. How do you reach "$\sqrt[3]{}$"? _____**

**70. How do you reach "I"? _____**

**71. How do you reach "x$^{-1}$"? _____**

**72. How do you reach "J"? _____**

**73. How do you reach "K"? _____**

**74. How do you reach "L"? _____**

**75. How do you reach "M"? _____**

**76. How do you reach "N"? _____**

**77. How do you reach "O"? _____**

**78. How do you reach "INS"? _____**

**79. How do you reach "OFF"? _____**

**80. How do you reach "CATALOG"? _____**

**81. How do you reach "P"? _____**

82. How do you reach “FORMAT”? _____

83. How do you reach “Q”? _____

84. How do you reach “R”? _____

85. How do you reach “{”? _____

86. How do you reach “S”? _____

87. How do you reach “}”? _____

88. How do you reach “T”? _____

89. How do you reach “List”? _____

90. How do you reach “U”? _____

91. How do you reach “Mat”? _____

92. How do you reach “V”? _____

93. How do you reach “W”? _____

94. How do you reach “[”? _____

95. How do you reach “X”? _____

96. How do you reach “]”? _____

97. How do you reach “Y”? _____

98. How do you reach “i”? _____

99. How do you reach “Z”? _____

100. How do you reach “=”? _____

101. How do you reach “π”? _____

**102. How do you reach “Ans”? _____**

**103. How do you reach “↵”? _____**

# Appendix A — Drill answers

## Set A

1. **55** (display `55`)

   #41 1 *(open the calculation app from the main menu)* → #42 2 → #37 5 → #44 + → #37 5 → #39 × → #38 6 → #50 EXE

2. **79** (display `79`)

   #41 1 *(open the calculation app from the main menu)* → #31 7 → #44 + → #32 8 → #39 × → #33 9 → #50 EXE

3. **282** (display `282`)

   #41 1 *(open the calculation app from the main menu)* → #41 1 → #37 5 → #46 0 → #44 + → #41 1 → #42 2 → #39 × → #41 1 → #41 1 → #50 EXE

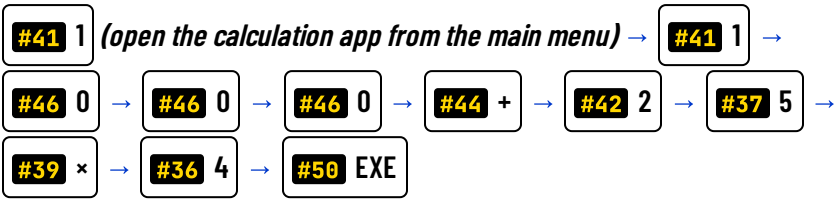
4. **213** (display `213`)

   #41 1 *(open the calculation app from the main menu)* → #43 3 → #44 + → #41 1 → #36 4 → #39 × → #41 1 → #37 5 → #50 EXE

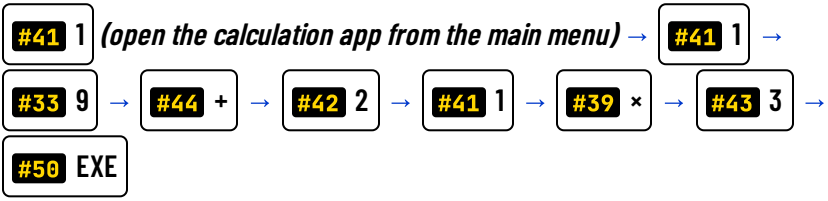
5. **90** (display `90`)

   #41 1 *(open the calculation app from the main menu)* → #36 4 → #32 8 → #44 + → #38 6 → #39 × → #31 7 → #50 EXE

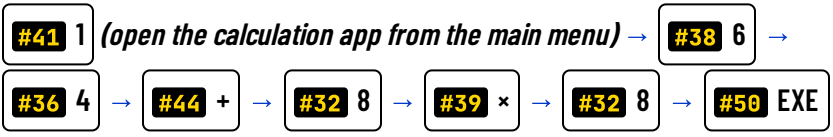
6. **1100** (display `1100`)



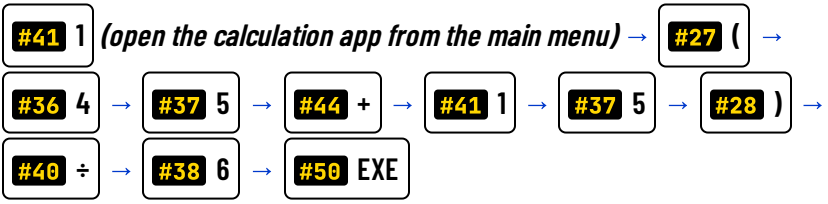

7. **82** (display `82`)



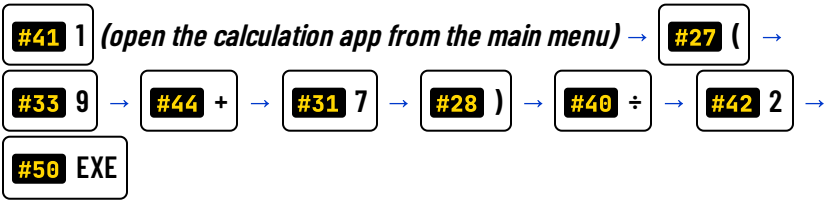

8. **128** (display `128`)




9. **10** (display `10`)




10. **8** (display `8`)



### 11. 20 (display 20)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#41 1 → #46 0 → #46 0 → #44 + → #38 6 → #46 0 →
#28 ) → #40 ÷ → #32 8 → #50 EXE

### 12. 10 (display 10)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#43 3 → #43 3 → #44 + → #41 1 → #31 7 → #28 ) →
#40 ÷ → #37 5 → #50 EXE

### 13. 50 (display 50)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#42 2 → #37 5 → #46 0 → #44 + → #43 3 → #37 5 →
#46 0 → #28 ) → #40 ÷ → #41 1 → #42 2 → #50 EXE

### 14. 77 (display 77)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#37 5 → #44 + → #38 6 → #28 ) → #39 × → #27 ( →
#41 1 → #46 0 → #45 − → #43 3 → #28 ) → #50 EXE

### 15. 45 (display 45)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#41 1 → #42 2 → #44 + → #43 3 → #28 ) → #39 × →
#27 ( → #32 8 → #45 − → #37 5 → #28 ) → #50 EXE

### 16. 12 (display 12)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#41 1 → #47 . → #37 5 → #44 + → #42 2 → #47 . →
#37 5 → #28 ) → #39 × → #27 ( → #31 7 → #45 − →
#36 4 → #28 ) → #50 EXE

### 17. 500 (display 500)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#42 2 → #46 0 → #44 + → #43 3 → #46 0 → #28 ) →
#39 × → #27 ( → #41 1 → #37 5 → #45 − → #37 5 →
#28 ) → #50 EXE

### 18. 990 (display 990)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#33 9 → #44 + → #41 1 → #28 ) → #39 × → #27 ( →
#41 1 → #46 0 → #46 0 → #45 − → #41 1 → #28 ) →
#50 EXE

### 19. 192 (display 192)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#41 1 → #41 1 → #44 + → #41 1 → #43 3 → #28 ) →
#39 × → #27 ( → #41 1 → #31 7 → #45 − → #33 9 →
#28 ) → #50 EXE

**20. 625** (display `625`)

`#41` 1 *(open the calculation app from the main menu)* → `#42` 2 → `#37` 5 → `#16` $x^2$ → `#50` EXE

**21. 49** (display `49`)

`#41` 1 *(open the calculation app from the main menu)* → `#31` 7 → `#16` $x^2$ → `#50` EXE

**22. 2.25** (display `2.25`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#47` . → `#37` 5 → `#16` $x^2$ → `#50` EXE

**23. 361** (display `361`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#33` 9 → `#16` $x^2$ → `#50` EXE

**24. 1089** (display `1089`)

`#41` 1 *(open the calculation app from the main menu)* → `#43` 3 → `#43` 3 → `#16` $x^2$ → `#50` EXE

**25. 0.25** (display `0.25`)

`#41` 1 *(open the calculation app from the main menu)* → `#46` 0 → `#47` . → `#37` 5 → `#16` $x^2$ → `#50` EXE

**26. 15625** (display `15625`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#42` 2 → `#37` 5 → `#16` $x^2$ → `#50` EXE

**27. 4096** (display `4096`)

`#41` 1 *(open the calculation app from the main menu)* → `#38` 6 → `#36` 4 → `#16` $x^2$ → `#50` EXE

**28. 25** (display `25`)

`#41` 1 *(open the calculation app from the main menu)* → `#7` SHIFT → `#16` √ → `#38` 6 → `#42` 2 → `#37` 5 → `#50` EXE

**29. 32** (display `32`)

`#41` 1 *(open the calculation app from the main menu)* → `#7` SHIFT → `#16` √ → `#41` 1 → `#46` 0 → `#42` 2 → `#36` 4 → `#50` EXE

**30. 99** (display `99`)

`#41` 1 *(open the calculation app from the main menu)* → `#7` SHIFT → `#16` √ → `#33` 9 → `#32` 8 → `#46` 0 → `#41` 1 → `#50` EXE

**31. 243** (display `243`)

`#41` 1 *(open the calculation app from the main menu)* → `#43` 3 → `#17` ^ → `#37` 5 → `#50` EXE

**32. 1.157625** (display `1.157625`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#47` . → `#46` 0 → `#37` 5 → `#17` ^ → `#43` 3 → `#50` EXE

**33. 625** (display `625`)

`#41` 1 *(open the calculation app from the main menu)* → `#37` 5 → `#17` ^ → `#36` 4 → `#50` EXE

**34. 343** (display `343`)

`#41` 1 *(open the calculation app from the main menu)* → `#31` 7 → `#17` ^ → `#43` 3 → `#50` EXE

**35. 2.25** (display `2.25`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#47` . → `#37` 5 → `#17` ^ → `#42` 2 → `#50` EXE

**36. 1000000.000** (display `1000000`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#46` 0 → `#17` ^ → `#38` 6 → `#50` EXE

**37. 1.41421356** (display `1.414213562`)

`#41` 1 *(open the calculation app from the main menu)* → `#42` 2 → `#17` ^ → `#46` 0 → `#47` . → `#37` 5 → `#50` EXE

**38. 5** (display `5`)

`#41` 1 *(open the calculation app from the main menu)* → `#46` 0 → `#47` . → `#42` 2 → `#7` SHIFT → `#28` $x^{-1}$ → `#50` EXE

**39. 2** (display `2`)

`#41` 1 *(open the calculation app from the main menu)* → `#20` log → `#41` 1 → `#46` 0 → `#46` 0 → `#28` ) → `#50` EXE

**40. 4** (display `4`)

`#41` 1 *(open the calculation app from the main menu)* → `#20` log → `#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#46` 0 → `#28` ) → `#50` EXE

**41. 2.30258509** (display `2.302585093`)

`#41` 1 *(open the calculation app from the main menu)* → `#21` ln → `#41` 1 → `#46` 0 → `#28` ) → `#50` EXE

**42. 0.69314718** (display `0.6931471806`)

`#41` 1 *(open the calculation app from the main menu)* → `#21` ln → `#42` 2 → `#28` ) → `#50` EXE

**43. -13** (display `−13`)

`#41` 1 *(open the calculation app from the main menu)* → `#49` (−) → `#42` 2 → `#46` 0 → `#44` + → `#31` 7 → `#50` EXE

**44. 0** (display `0`)

`#41` 1 *(open the calculation app from the main menu)* → `#49` (−) → `#43` 3 → `#44` + → `#43` 3 → `#50` EXE

**45. 150** (display `150`)

`#41` 1 *(open the calculation app from the main menu)* → `#49` (−) → `#41` 1 → `#46` 0 → `#46` 0 → `#44` + → `#42` 2 → `#37` 5 → `#46` 0 → `#50` EXE

**46. -7** (display `−7`)

`#41` 1 *(open the calculation app from the main menu)* → `#49` (−) → `#32` 8 → `#44` + → `#41` 1 → `#50` EXE

**47. 8000** (display `8000`)

`#41` 1 *(open the calculation app from the main menu)* → `#42` 2 → `#48` EXP → `#43` 3 → `#39` × → `#36` 4 → `#50` EXE

**48. 30000** (display `30000`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#47` . → `#37` 5 → `#48` EXP → `#36` 4 → `#39` × → `#42` 2 → `#50` EXE

**49. 900000** (display `900000`)

`#41` 1 *(open the calculation app from the main menu)* → `#43` 3 → `#48` EXP → `#37` 5 → `#39` × → `#43` 3 → `#50` EXE

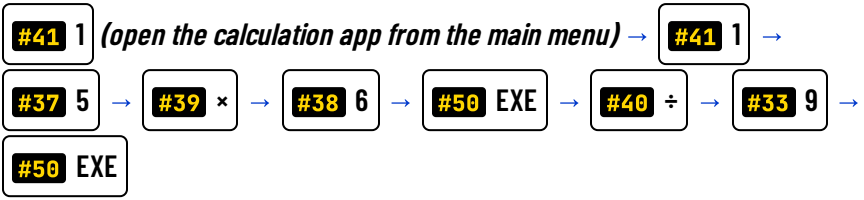
**50. 900** (display `900`)

`#41` 1 *(open the calculation app from the main menu)* → `#36` 4 → `#47` . → `#37` 5 → `#48` EXP → `#42` 2 → `#39` × → `#42` 2 → `#50` EXE

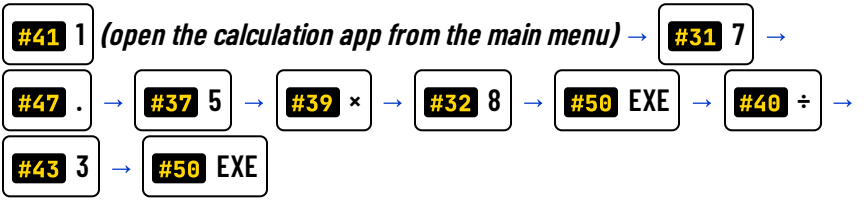
**51. 4.52389342** (display `4.523893421`)

`#41` 1 *(open the calculation app from the main menu)* → `#7` SHIFT → `#48` π → `#39` × → `#41` 1 → `#47` . → `#42` 2 → `#16` $x^2$ → `#50` EXE

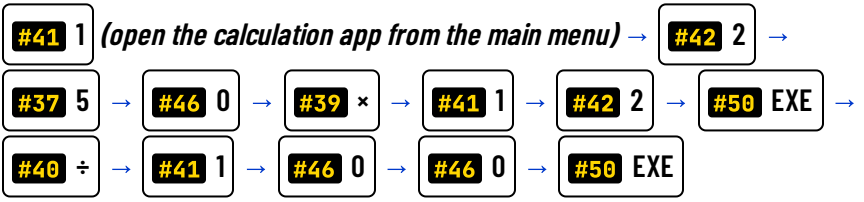
**52. 10** (display `10`)



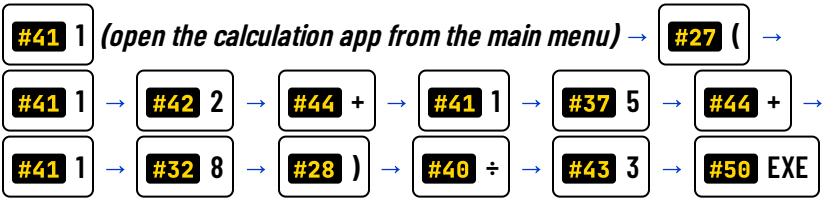

**53. 20** (display `20`)



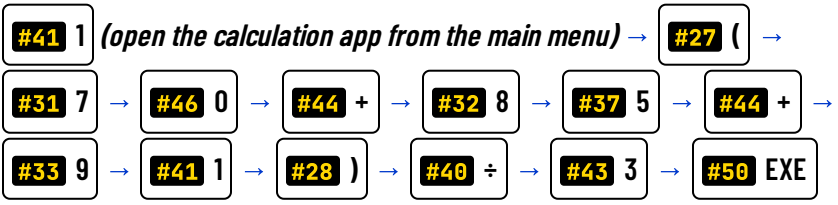

**54. 30** (display `30`)



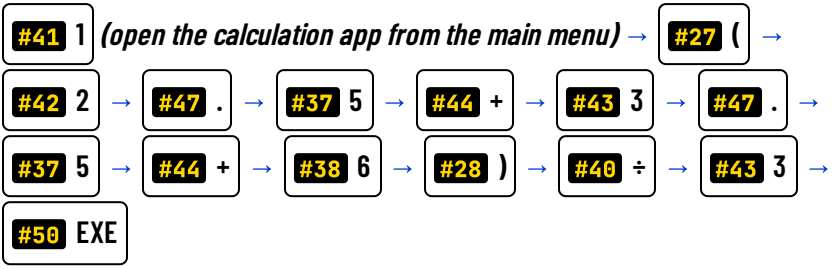

**55. 15** (display `15`)



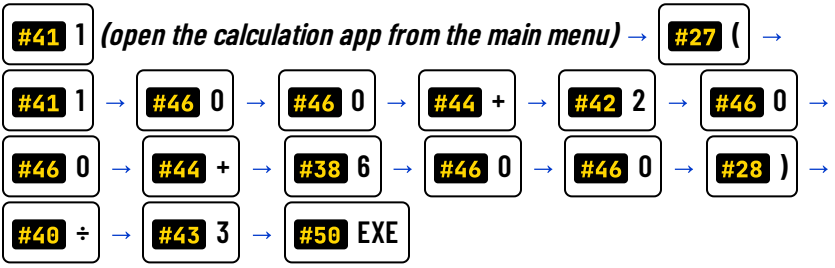

**56. 82** (display `82`)



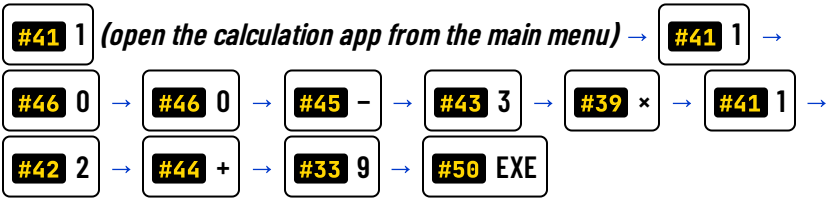
**57. 4** (display `4`)



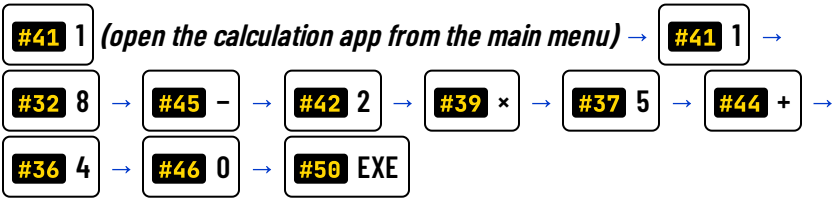

**58. 300** (display `300`)




**59. 73** (display `73`)




**60. 48** (display `48`)



### 61. 5 (display 5)

#41 1 *(open the calculation app from the main menu)* → #31 7 → #47 . → #37 5 → #45 − → #41 1 → #47 . → #37 5 → #39 × → #43 3 → #44 + → #42 2 → #50 EXE

### 62. 42 (display 42)

#41 1 *(open the calculation app from the main menu)* → #38 6 → #36 4 → #45 − → #32 8 → #39 × → #36 4 → #44 + → #41 1 → #46 0 → #50 EXE

### 63. 255 (display 255)

#41 1 *(open the calculation app from the main menu)* → #43 3 → #46 0 → #46 0 → #45 − → #42 2 → #37 5 → #39 × → #36 4 → #44 + → #37 5 → #37 5 → #50 EXE

### 64. 1 (display 1)

#41 1 *(open the calculation app from the main menu)* → #33 9 → #45 − → #43 3 → #39 × → #43 3 → #44 + → #41 1 → #50 EXE

### 65. 35 (display 35)

#41 1 *(open the calculation app from the main menu)* → #36 4 → #37 5 → #45 − → #37 5 → #39 × → #37 5 → #44 + → #41 1 → #37 5 → #50 EXE

**66. 25** (display `25`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#40` ÷ → `#32` 8 → `#40` ÷ → `#37` 5 → `#50` EXE

**67. 16** (display `16`)

`#41` 1 *(open the calculation app from the main menu)* → `#33` 9 → `#38` 6 → `#40` ÷ → `#42` 2 → `#40` ÷ → `#43` 3 → `#50` EXE

**68. 2** (display `2`)

`#41` 1 *(open the calculation app from the main menu)* → `#31` 7 → `#42` 2 → `#46` 0 → `#46` 0 → `#40` ÷ → `#38` 6 → `#46` 0 → `#40` ÷ → `#38` 6 → `#46` 0 → `#50` EXE

**69. 5** (display `5`)

`#41` 1 *(open the calculation app from the main menu)* → `#37` 5 → `#46` 0 → `#46` 0 → `#40` ÷ → `#36` 4 → `#40` ÷ → `#42` 2 → `#37` 5 → `#50` EXE

**70. 9** (display `9`)

`#41` 1 *(open the calculation app from the main menu)* → `#32` 8 → `#41` 1 → `#40` ÷ → `#43` 3 → `#40` ÷ → `#43` 3 → `#50` EXE

### 71. 5 (display 5)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#27 ( → #41 1 → #46 0 → #44 + → #42 2 → #28 ) →
#39 × → #43 3 → #45 − → #38 6 → #28 ) → #40 ÷ →
#38 6 → #50 EXE

### 72. 4 (display 4)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#27 ( → #41 1 → #47 . → #37 5 → #44 + → #42 2 →
#47 . → #37 5 → #28 ) → #39 × → #38 6 → #45 − →
#36 4 → #28 ) → #40 ÷ → #37 5 → #50 EXE

### 73. 25 (display 25)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#27 ( → #42 2 → #46 0 → #44 + → #43 3 → #46 0 →
#28 ) → #39 × → #42 2 → #45 − → #42 2 → #37 5 →
#28 ) → #40 ÷ → #43 3 → #50 EXE

### 74. 10 (display 10)

#41 1 *(open the calculation app from the main menu)* → #27 ( →
#27 ( → #31 7 → #44 + → #32 8 → #28 ) → #39 × →
#33 9 → #45 − → #41 1 → #37 5 → #28 ) → #40 ÷ →
#41 1 → #42 2 → #50 EXE

**75. 24** (display `24`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#42` 2 → `#44` + → `#43` 3 → `#39` × → `#36` 4 → `#50` EXE

**76. 5** (display `5`)

`#41` 1 *(open the calculation app from the main menu)* → `#27` ( → `#41` 1 → `#42` 2 → `#44` + → `#32` 8 → `#28` ) → `#40` ÷ → `#36` 4 → `#50` EXE

**77. 49** (display `49`)

`#41` 1 *(open the calculation app from the main menu)* → `#27` ( → `#43` 3 → `#44` + → `#36` 4 → `#28` ) → `#39` × → `#27` ( → `#33` 9 → `#45` − → `#42` 2 → `#28` ) → `#50` EXE

**78. 144** (display `144`)

`#41` 1 *(open the calculation app from the main menu)* → `#41` 1 → `#42` 2 → `#16` $x^2$ → `#50` EXE

**79. 12** (display `12`)

`#41` 1 *(open the calculation app from the main menu)* → `#7` SHIFT → `#16` √ → `#41` 1 → `#36` 4 → `#36` 4 → `#50` EXE

**80. 1024** (display `1024`)

`#41` 1 *(open the calculation app from the main menu)* → `#42` 2 → `#17` ^ → `#41` 1 → `#46` 0 → `#50` EXE

**81. 4 (display `4`)**

`#41` 1 *(open the calculation app from the main menu)* → `#46` 0 → `#47` . → `#42` 2 → `#37` 5 → `#7` SHIFT → `#28` $x^{-1}$ → `#50` EXE

**82. 1 (display `1`)**

`#41` 1 *(open the calculation app from the main menu)* → `#23` cos → `#46` 0 → `#28` ) → `#50` EXE

**83. 3 (display `3`)**

`#41` 1 *(open the calculation app from the main menu)* → `#20` log → `#41` 1 → `#46` 0 → `#46` 0 → `#46` 0 → `#28` ) → `#50` EXE

**84. 0 (display `0`)**

`#41` 1 *(open the calculation app from the main menu)* → `#21` ln → `#41` 1 → `#28` ) → `#50` EXE

**85. 7 (display `7`)**

`#41` 1 *(open the calculation app from the main menu)* → `#49` (−) → `#37` 5 → `#44` + → `#41` 1 → `#42` 2 → `#50` EXE

**86. 1800 (display `1800`)**

`#41` 1 *(open the calculation app from the main menu)* → `#38` 6 → `#48` EXP → `#42` 2 → `#39` × → `#43` 3 → `#50` EXE

**87. 19.63495408** (display `19.63495408`)

#41 1 *(open the calculation app from the main menu)* → #7 SHIFT → #48 π → #39 × → #42 2 → #47 . → #37 5 → #16 $x^2$ → #50 EXE

**88. 200** (display `200`)

#41 1 *(open the calculation app from the main menu)* → #41 1 → #46 0 → #46 0 → #39 × → #36 4 → #50 EXE → #40 ÷ → #42 2 → #50 EXE

**89. 9** (display `9`)

#41 1 *(open the calculation app from the main menu)* → #27 ( → #36 4 → #44 + → #32 8 → #44 + → #41 1 → #37 5 → #28 ) → #40 ÷ → #43 3 → #50 EXE

**90. 33** (display `33`)

#41 1 *(open the calculation app from the main menu)* → #37 5 → #46 0 → #45 − → #36 4 → #39 × → #38 6 → #44 + → #31 7 → #50 EXE

**91. 10** (display `10`)

#41 1 *(open the calculation app from the main menu)* → #43 3 → #38 6 → #46 0 → #40 ÷ → #36 4 → #40 ÷ → #33 9 → #50 EXE

**92. 3 (display 3)**

#41 1 *(open the calculation app from the main menu)* → #27 ( → #27 ( → #43 3 → #44 + → #37 5 → #28 ) → #39 × → #36 4 → #45 − → #42 2 → #28 ) → #40 ÷ → #41 1 → #46 0 → #50 EXE

## Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #34
32. #35
33. #39
34. #40
35. #44
36. #45
37. #47
38. #48
39. #49
40. #50
41. #7 SHIFT then #1 F1
42. #7 SHIFT then #2 F2
43. #7 SHIFT then #3 F3
44. #7 SHIFT then #4 F4
45. #7 SHIFT then #5 F5
46. #7 SHIFT then #6 F6
47. #7 SHIFT then #9 VARS
48. #7 SHIFT then #10 MENU
49. #7 SHIFT then #15 ALPHA
50. #7 SHIFT then #16 $x^2$
51. #15 ALPHA then #16 $x^2$
52. #7 SHIFT then #17 ^
53. #15 ALPHA then #17 ^
54. #7 SHIFT then #18 EXIT
55. #15 ALPHA then #19 X,θ,T
56. #7 SHIFT then #20 log
57. #15 ALPHA then #20 log
58. #7 SHIFT then #21 ln

59. #15 ALPHA then #21 ln
60. #7 SHIFT then #22 sin
61. #15 ALPHA then #22 sin
62. #7 SHIFT then #23 cos
63. #15 ALPHA then #23 cos
64. #7 SHIFT then #24 tan
65. #15 ALPHA then #24 tan
66. #7 SHIFT then #25 a/b
67. #15 ALPHA then #25 a/b
68. #15 ALPHA then #26 F↔D
69. #7 SHIFT then #27 (
70. #15 ALPHA then #27 (
71. #7 SHIFT then #28 )
72. #15 ALPHA then #28 )
73. #15 ALPHA then #29 ,
74. #15 ALPHA then #30 →
75. #15 ALPHA then #31 7
76. #15 ALPHA then #32 8
77. #15 ALPHA then #33 9
78. #7 SHIFT then #34 DEL
79. #7 SHIFT then #35 AC/ON
80. #7 SHIFT then #36 4
81. #15 ALPHA then #36 4
82. #7 SHIFT then #37 5
83. #15 ALPHA then #37 5
84. #15 ALPHA then #38 6
85. #7 SHIFT then #39 ×
86. #15 ALPHA then #39 ×
87. #7 SHIFT then #40 ÷
88. #15 ALPHA then #40 ÷
89. #7 SHIFT then #41 1
90. #15 ALPHA then #41 1
91. #7 SHIFT then #42 2
92. #15 ALPHA then #42 2
93. #15 ALPHA then #43 3
94. #7 SHIFT then #44 +
95. #15 ALPHA then #44 +
96. #7 SHIFT then #45 -
97. #15 ALPHA then #45 -
98. #7 SHIFT then #46 0
99. #15 ALPHA then #46 0
100. #7 SHIFT then #47 .
101. #7 SHIFT then #48 EXP
102. #7 SHIFT then #49 (-)
103. #7 SHIFT then #50 EXE

# Appendix B — Function finder (A–Z)

| Function | Label | Keys |
|---|---|---|
| ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23 | EXP | #48 |
| 10ˣ — antilog | 10x | #7 → #20 |
| Addition + | + | #44 |
| ALPHA — the next key types its letter/variable (red/green label) | ALPHA | #15 |
| ALPHA — the next key types its letter/variable (red/green label) | A-LOCK | #7 → #15 |
| Ans — recall the last answer into the current expression | Ans | #7 → #49 |
| CALC — graph analysis (zero, min, max, intersect) | G-Solv | #7 → #5 |
| CATALOG — list of all commands | CATALOG | #7 → #36 |
| Closing bracket ). Closing brackets at the end of a line may be omitted | ) | #28 |
| Comma / argument separator (between function arguments, list items or coordinates) | , | #29 |
| Common logarithm $\log_{10}$ | log | #20 |
| Cosine (current angle unit) | cos | #23 |
| Cube root $\sqrt[3]{}$ | $\sqrt[3]{}$ | #7 → #27 |
| Cursor ▲ — up / previous history line / previous item | ▲ | #11 |
| Cursor ▶ — move right in the line | ▶ | #14 |
| Cursor ▼ — down / next history line / next item | ▼ | #12 |
| Cursor ◀ — move left in the line | ◀ | #13 |
| Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored | . | #47 |
| DEL — delete the character at the cursor | DEL | #34 |
| Digit 0 | 0 | #46 |

| | | |
|---|---|---|
| Digit 1 | 1 | #41 |
| Digit 2 | 2 | #42 |
| Digit 3 | 3 | #43 |
| Digit 4 | 4 | #36 |
| Digit 5 | 5 | #37 |
| Digit 6 | 6 | #38 |
| Digit 7 | 7 | #31 |
| Digit 8 | 8 | #32 |
| Digit 9 | 9 | #33 |
| Division ÷ | ÷ | #40 |
| DRAW menu | Sketch | #7 → #4 |
| Equals — complete the calculation | = | #7 → #47 |
| $e^x$ — exponential | ex | #7 → #21 |
| Execute / = — evaluate the line and show the result (EXE / ENTER / =) | EXE | #50 |
| Execute / = — evaluate the line and show the result (EXE / ENTER / =) | ↵ | #7 → #50 |
| F1 soft key | F1 | #1 |
| F2 soft key | F2 | #2 |
| F3 soft key | F3 | #3 |
| F4 soft key | F4 | #4 |
| F5 soft key | F5 | #5 |
| F6 soft key | F6 | #6 |
| FORMAT — choose number format (Fix/Sci/Norm) | FORMAT | #7 → #37 |
| Fraction template ▭/▭ (numerator over denominator) | a⁄b | #25 |
| Graph / app mode | SET UP | #7 → #10 |

| | | |
|---|---|---|
| Imaginary unit i (complex numbers) | i | #7 → #46 |
| INS — insert mode at the cursor | INS | #7 → #34 |
| Inverse cosine $\cos^{-1}$ | $\cos^{-1}$ | #7 → #23 |
| Inverse sine $\sin^{-1}$ | $\sin^{-1}$ | #7 → #22 |
| Inverse tangent $\tan^{-1}$ | $\tan^{-1}$ | #7 → #24 |
| LIST menu | List | #7 → #41 |
| MATRIX menu | Mat | #7 → #42 |
| MENU — main menu | MENU | #10 |
| Mixed-number template (whole, numerator, denominator) | □/□ | #7 → #25 |
| Multiplication × | × | #39 |
| n-th root $\sqrt[x]{y}$ | $x\sqrt{}$ | #7 → #17 |
| Natural logarithm ln | ln | #21 |
| Negative sign (−) — makes the next number negative. Different from the subtraction key | (−) | #49 |
| OFF — power off (state is lost; press ON to restart) | OFF | #7 → #35 |
| ON / AC — power on, or clear all | AC/ON | #35 |
| Opening bracket ( | ( | #27 |
| OPTN — options for the current mode | OPTN | #8 |
| Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512) | ^ | #17 |
| PRGM — programs | PRGM | #7 → #9 |
| QUIT — leave the current screen / app | EXIT | #18 |

| | | |
|---|---|---|
| QUIT — leave the current screen / app | QUIT | #7 → #18 |
| Reciprocal $x^{-1}$ (1/x) | $x^{-1}$ | #7 → #28 |
| S⇔D — toggle the result between exact (fraction, √, π) and decimal | F ↔ D | #26 |
| SHIFT / 2nd — the next key uses its upper (shift-coloured) function | SHIFT | #7 |
| Sine (uses the current angle unit) | sin | #22 |
| Square root √ | √ | #7 → #16 |
| Square $x^2$ | $x^2$ | #16 |
| STO▶ ( → ) — store into a variable | → | #30 |
| Subtraction – | – | #45 |
| TABLE — table of values | G ↔ T | #7 → #6 |
| Tangent (current angle unit) | tan | #24 |
| The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving | X,θ,T | #19 |
| TRACE — move along a graph | Trace | #7 → #1 |
| Types "[" | [ | #7 → #44 |
| Types "]" | ] | #7 → #45 |
| Types "{" | { | #7 → #39 |
| Types "}" | } | #7 → #40 |
| V-Window — graph view settings | V-Window | #7 → #3 |
| Variable A (store with STO/ →, recall by name) | A | #15 → #19 |
| Variable B (store with STO/ →, recall by name) | B | #15 → #20 |

| | | |
|---|---|---|
| Variable C (store with STO/→, recall by name) | C | #15 → #21 |
| Variable D (store with STO/→, recall by name) | D | #15 → #22 |
| Variable E (store with STO/→, recall by name) | E | #15 → #23 |
| Variable F (store with STO/→, recall by name) | F | #15 → #24 |
| Variable G (store with STO/→, recall by name) | G | #15 → #25 |
| Variable H (store with STO/→, recall by name) | H | #15 → #26 |
| Variable I (store with STO/→, recall by name) | I | #15 → #27 |
| Variable J (store with STO/→, recall by name) | J | #15 → #28 |
| Variable K (store with STO/→, recall by name) | K | #15 → #29 |
| Variable L (store with STO/→, recall by name) | L | #15 → #30 |
| Variable M (store with STO/→, recall by name) | M | #15 → #31 |
| Variable N (store with STO/→, recall by name) | N | #15 → #32 |
| Variable O (store with STO/→, recall by name) | O | #15 → #33 |
| Variable P (store with STO/→, recall by name) | P | #15 → #36 |
| Variable Q (store with STO/→, recall by name) | Q | #15 → #37 |
| Variable r (store with STO/→, recall by name) | r | #15 → #16 |

| | | |
|---|---|---|
| Variable R (store with STO/→, recall by name) | R | #15 → #38 |
| Variable S (store with STO/→, recall by name) | S | #15 → #39 |
| Variable T (store with STO/→, recall by name) | T | #15 → #40 |
| Variable U (store with STO/→, recall by name) | U | #15 → #41 |
| Variable V (store with STO/→, recall by name) | V | #15 → #42 |
| Variable W (store with STO/→, recall by name) | W | #15 → #43 |
| Variable X (store with STO/→, recall by name) | X | #15 → #44 |
| Variable Y (store with STO/→, recall by name) | Y | #15 → #45 |
| Variable Z (store with STO/→, recall by name) | Z | #15 → #46 |
| Variable θ (store with STO/→, recall by name) | θ | #15 → #17 |
| VARS — variables menu | VARS | #9 |
| ZOOM — zoom presets | Zoom | #7 → #2 |
| π (3.14159265…) as an exact constant | π | #7 → #48 |

# Appendix C — Glossary

**Ans**
The result of the last calculation, reusable in the next.

**AOS**
Algebraic Operating System — operator precedence on a one-line display.

**RPN**
Reverse Polish Notation — operands first, then the operation; no = key.

**Stack**
The RPN registers X, Y, Z, T where numbers wait for operations.

**TVM**
Time Value of Money — n, i, PV, PMT, FV.

**Natural display**
Shows fractions, roots and powers as printed in textbooks.

**Shift / 2nd**
Prefix key that selects the function printed above a key.

**Alpha**
Prefix key that types the letter printed beside a key.

**Radian**
Angle unit; 2π rad = 360°.

**Grad**
Angle unit; 400 grad = 360°.

**Engineering notation**
Scientific notation with exponents that are multiples of 3.

**Mantissa**
The digits of a number in scientific notation.

**Memory (M)**

An independent register for running totals.

**Mark-up**

Increase over cost; margin is profit as a percent of selling price.

**Grand total (GT)**

Sum of all totals since the last clear.

**nCr / nPr**

Combinations / permutations of n items taken r at a time.

**Factorial**

n! = 1×2×…×n.

**Reciprocal**

1/x.

**Standard deviation**

Spread of data around the mean (σ population, s sample).

**Key number**

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

# Appendix D — Model notes & limits

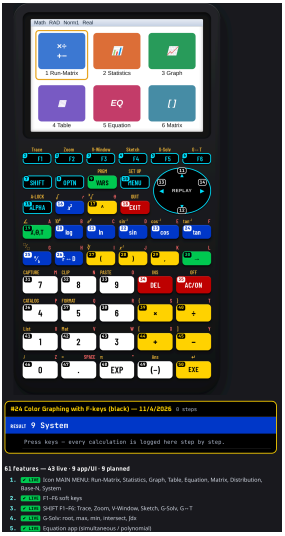
F-key graphing model. MENU opens the icon menu (Run-Matrix, Statistics, Graph, Table, Equation, Matrix, Distribution, Base-N, System) – pick with the cursor + EXE or the number. F1–F6 act on the soft-key bar; SHIFT F1–F6 give Trace, Zoom, V-Window, Sketch, G-Solv (root, max, min, intersect, ∫dx) and G ↔ T. Colour screen with natural input. Programs, e-Activity, spreadsheet, CAPTURE/CLIP/PASTE are not simulated.

## Planned (not yet working)

- Programs
- e-Activity
- Spreadsheet
- CAPTURE / CLIP / PASTE
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

# Appendix E — Quick-reference card




| | | |
|---|---|---|
| **#1** F1 ⇧ Trace | **#2** F2 ⇧ Zoom | **#3** F3 ⇧ V-Window |
| **#4** F4 ⇧ Sketch | **#5** F5 ⇧ G-Solv | **#6** F6 ⇧ G↔T |
| **#7** SHIFT | **#8** OPTN | **#9** VARS ⇧ PRGM |
| **#10** MENU ⇧ SET UP | **#11** ▲ | **#12** ▼ |
| **#13** ◀ | **#14** ▶ | **#15** ALPHA ⇧ A-LOCK |
| **#16** $x^2$ ⇧ √ | **#17** ^ ⇧ x√ | **#18** EXIT ⇧ QUIT |
| **#19** X,θ,T ⇧ ∠ | **#20** log ⇧ 10x | **#21** ln ⇧ $e^x$ |
| **#22** sin ⇧ $\sin^{-1}$ | **#23** cos ⇧ $\cos^{-1}$ | **#24** tan ⇧ $\tan^{-1}$ |
| **#25** a⁄b ⇧ □⁄□ | **#26** F ↔ D | **#27** ( ⇧ $\sqrt[3]{}$ |
| **#28** ) ⇧ $x^{-1}$ | **#29** , | **#30** → |
| **#31** 7 ⇧ CAPTURE | **#32** 8 ⇧ CLIP | **#33** 9 ⇧ PASTE |
| **#34** DEL ⇧ INS | **#35** AC/ON ⇧ OFF | **#36** 4 ⇧ CATALOG |
| **#37** 5 ⇧ FORMAT | **#38** 6 | **#39** × ⇧ { |
| **#40** ÷ ⇧ } | **#41** 1 ⇧ List | **#42** 2 ⇧ Mat |
| **#43** 3 | **#44** + ⇧ [ | **#45** − ⇧ ] |
| **#46** 0 ⇧ i | **#47** . ⇧ = | **#48** EXP ⇧ π |
| **#49** (−) ⇧ Ans | **#50** EXE ⇧ ↵ | |

U-45 CALCULATORS · Calculator #24 · cut out and keep